

“Ask yourself WHY you want to track yourself”: Co-Designing a Self-Tracking Guide with Teenagers

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ABSTRACT

Teenagers often engage with Personal Informatics tools for health and fitness without support or guidance specific to their individual experiences. There is a lack of research on how putting teens at the center of designing educational resources might empower young users to harness the affordances of self-tracking tools that are of most value to them. To address this gap, we ran five online co-design workshops with 44 teenagers (aged 13–18 years) in the United Kingdom to design a self-tracking guide for teens. We focused on content, presentation of information, and distribution strategy. Our findings highlight the complexity of creating resources which adeptly capture the needs of this diverse user group. To successfully build interactive educational resources appropriate for teens’ positive engagement with health-related self-tracking tools, designers and researchers must prioritize teen-centered design with a focus on: (1) teens’ core reasons for self-tracking and (2) opportunities for social support and community-building.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile computing design and evaluation methods; Human computer interaction (HCI); HCI design and evaluation methods; • **Applied computing** → Life and medical sciences; Health informatics.

KEYWORDS

Self-tracking, personal informatics, co-design workshops, youth, education

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1 INTRODUCTION

Personal informatics (PI) systems, such as commercial self-tracking devices and apps, allow users to gather and reflect on personal data, including health data such as daily step counts and calories burned

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[13]. A recent report in the United Kingdom found that over half of teenagers reported using self-tracking tools, often doing so without a clear understanding of what the data means [23]. This raises urgent questions around how we can develop resources which might best support young users of these PI tools.

The teenage years are a critical developmental period for life-long health behaviors [10, 16]; however, minimal work has been done to empower teen users to engage with PI tools to track their health and fitness in healthy, personally relevant ways. Teens express uncertainty about how to best harness the affordances of digital technologies for health for their individual circumstances [23], and they often lack the social support and educational resources required to access appropriate health information and feel confident in making informed health decisions [4, 8, 23].

Previous work has recognized that teenagers’ engagement with PI tools might look different to adults’ [5, 18, 20]. Nevertheless, little has been done to explore how teens might imagine the design of educational resources around these tools differently to fit their purposes. In this late breaking work, we attempt to address this gap by working with teenagers (aged 13–18 years) in online workshops to co-design a reimagined ‘user guide’ for individuals interested in self-tracking, exploring design considerations for the content, format, and distribution of such a resource.

Teenagers are still missed in much interaction design and product design work [3, 11, 17, 22]. The problem here, as Spiel et al. [25] emphasize, is that rigid, normative design reflects the goals and ideals of designers, rather than the diverse practices of the end users who might engage with these tools. The result is that teenagers often feel frustrated that current PI tools are not designed with their age group in mind [2, 6].

2 RELATED WORK

Recent research in Human-Computer Interaction (HCI) has sought to address the gap in our understanding of teens’ self-tracking by engaging with teenagers through innovative extended co-design and participatory design methods to delve into their everyday self-tracking experiences [18, 19]. This work has highlighted how we might think about teenagers’ reasons for self-tracking and the effects of their tracking practices differently from adults’. For example, this research emphasizes the importance of affordances of flexibility and self-expression, and the ways in which these practices can foster social connection, learning about the self, and identity development [19]. Further, privacy and curation affordances are key for this population [19] and teenagers are concerned about how their tracking allows them to fit in with accepted social norms, which are often set by adults [18]. More research is needed to explore how additional educational resources designed specifically for this

population might empower teens to harness the unique affordances of existing PI tools that are of most value to them.

Importantly, despite teens' desire to be included in the design and regulation of digital technologies for health [23], no work critically explores what teens understand about self-tracking and need from educational resources. Co-design methods have previously been used with teenagers to inform the design of educational health resources [14]. As such, the aim of this study was to work alongside teenagers towards designing a teen-centered user guide (i.e., a 'how-to' manual for effectively using PI tools written with and for teens) to be developed for those who might be interested in self-tracking. We chose to use the term 'user guide' in our workshops to reflect that our focus remained firmly on teenagers as core users. We found teens were familiar with the term 'user guide', and this helped them co-design creative solutions in our activities. We oriented away from terms such as 'educational resource' or 'design guide' as we were not engaging with professional designers or educators. Through co-design workshops, we collaborated with teens to uncover topics they felt were most important and relevant to their age group, as well as when and how they felt information about self-tracking should be shared with their peers. In this work, we find that foregrounding individual reasons for self-tracking and opportunities for social support might pave the way for designing teen-centered resources for self-tracking.

3 METHODS

For the purpose of this work, the scope of self-tracking was confined to PI for health and fitness. This paper reports on five online co-design workshops, which formed part of a larger mixed-methods study involving around 800 teenagers (aged 13-18 years) in the United Kingdom to explore their self-tracking practices during the COVID-19 pandemic and beyond. All five workshops were conducted in November 2021.

3.1 Participants

Participants were recruited through a number of channels: those already involved in other research activities in this larger project (i.e., online survey and interviews), social media posts, advertisements distributed by youth organizations, and the researchers' personal networks. Participants were compensated with a £15 online shopping voucher for their participation.

Parental consent and participant assent was obtained online for participants under the age of 16. For those over the age of 16, online participant consent was given. Before any data collection commenced, this study was granted ethics approval by the University of Oxford Central University Research Ethics Committee (CUREC).

Seventy-six teenagers signed up to participate, 44 of whom successfully completed a workshop (13 identified as female, 20 as male, and 11 did not disclose). Of these 44 teenagers, 23 were currently using self-tracking tools, 10 had previously used self-tracking tools, three had never used self-tracking tools, and eight did not disclose their experience of self-tracking.

3.2 Procedure

We ran five 90-minute online workshops with 4-15 participants in each session¹. The workshops took place using the videoconferencing software *Microsoft Teams* and the online whiteboard platform *Miro*. Participants were able to use the chat, video, and audio functions in *Microsoft Teams* to participate however they were most comfortable.

The workshops consisted of three key activities, besides an introductory activity (familiarizing participants with the online whiteboard functions and each other):

- **Persona exploration:** working with personas (defined by the larger study) to uncover a variety of teenage experiences with PI tools (including reasons for tracking, frequency of engagement, opportunities, challenges)
- **Feature evaluation:** identifying key features of PI tools and their effects to understand pain points and opportunities for improvement (including features of interest, features of concern, and social and emotional effects of PI tools)
- **Designing a teen-centered 'user guide':** expanding on Activities 1 and 2 to identify key design considerations for a user guide (content it should include, how it should be formatted, and how it should be shared)

In this late breaking work, we focus closely on the third activity focused on designing a teen-centered user guide as this was the culmination of all of the workshop activities. The workshops were video-recorded and transcribed verbatim. Transcripts, *Microsoft Teams* chat histories, and *Miro* boards were analyzed together by two researchers using thematic analysis [1]. Participant quotations and contributions on the boards are reported verbatim (including any spelling or grammatical errors) to retain the authenticity of the teens' voices. Participants are identified by workshop number only (i.e., W2 denotes a participant from Workshop 2).

4 FINDINGS

We have organized our findings into three critical considerations when designing a user guide: (1) key content, (2) information format, and (3) distribution strategy. An example of a *Miro* board is available in Figure 1.

4.1 What Key Content Would Teenagers Like to Be Included in a Self-Tracking User Guide?

Participants focused on five themes related to key content for a user guide: (1) acknowledging individual experiences, (2) caution around obsessive behaviors, (3) self-tracking as community-building, (4) importance of context in self-tracked data, and (5) balancing privacy and the need for tailored support. All categories were discussed during each workshop.

4.1.1 Why Self-Tracking? Acknowledging Individual Experiences. Teenagers were cognizant of differing self-tracking experiences and acknowledged the importance of discussing these tensions in a user guide. For some, self-tracking was an important part of their

¹We wanted to ensure we did not compromise participants' anonymity on the whiteboards by having too few participants in each workshop. As such, we decided to oversample for each workshop, thus some workshops had more participants than others. We created 'break out' rooms in *Microsoft Teams* if workshops were too large; however, these were not required.



Figure 1: A screenshot of Activity 3 on the Miro board from Workshop 4, demonstrating participants’ use of sticky notes, emojis, and arrows to connect ideas. This was supplemented by verbal discussion and comments using the ‘Chat’ function.

routine and self-care; others felt the use of PI tools was, or could be, detrimental to their psychological wellbeing. Critically, participants addressed the importance of different reasons for self-tracking in understanding individual experiences: “Ask yourself WHY you want to track yourself - if not positive then don’t do it!” (W4).

For example, this question of “WHY” was prominent when discussing calorie tracking. Some participants highlighted the benefits of calorie tracking for specific individuals, such as for teenage athletes, whilst others were staunchly against the practice: “if someone asked me if they should start tracking calories my answer would be a big resounding NO” (W2). Despite differences of opinion, participants were clear they wanted the user guide to encourage nuanced, “not judgmental” (W2) conversations about this. As one participant stated, “Tracking is for everyone and also not for everyone. It’s helpful for people who can bear listening or knowing about their health status” (W3). Participants wanted the user guide to be unafraid to address these tensions.

Participants agreed it is not only teenagers who needed to be better educated. They argued that a user guide must expand this non-judgmental space to include parents and schools as well. The participants articulated that teenagers, educators, and parents need to be aware of individual reasons for engaging in self-tracking and the nuanced experiences, both positive and negative, that come alongside those.

4.1.2 How to Self-Track? Caution Around Obsessive Behaviors. Participants repeatedly used cautionary language when describing

self-tracking, including phrases such as “be careful” (W5). The teens acknowledged the benefits of self-tracking, but many were concerned about it being a slippery slope: “it is so easy to become obsessed with” (W5). Some participants wanted to warn others about the dangers of tracking beyond a certain threshold of use: “don’t get too drawn in” (W1) and “keep an eye out for obsessive tracking” (W5).

For some, the dangers of obsession were related to the reasons for using PI tools: “i think that self tracking can be good for competition with peers, as in fitness goals and prepping for competitive sport, however it can definitely effect you negatively if you become obsessed with your aims, especially if they’re unhealthy obsessions/habits” (W5). Some also referenced the role of following social media influencers in prompting obsessive tracking behaviors.

Short of defining when too much is too much, participants agreed that each individual should determine their own balance: “Obviously when pushed to extremes its negative, but as long as you can use your self-awareness to decide if you’re going too far then you should be fine” (W4). Some stressed that community can be important for countering obsessive habits: “track with friends and family who will support you when you are doing well, but also inform you when you are doing too much and becoming unhealthy, physically or mentally” (W1). The teens felt the user guide must discuss the dangers of obsessive tracking and provide resources for identifying and addressing such habits, such as fostering supportive communities.

4.1.3 Sharing the Tracking Experience? Self-Tracking as Community-Building. Teens agreed that community-building is important for healthy self-tracking practices and suggested that the user guide could promote this community orientation. For some, the idea of community involved competitions. Others simply imagined a space for conversation, support, and care: *“The community shouldn’t just be competitive. it should be a community where you can relate easily with other people in the community”* (W3).

To facilitate healthy communities, participants discussed moderators and the absence of leaderboards. Moderators—whether third parties, older students, or peers—were contentious, with no consensus reached. Participants broadly agreed that leaderboards were more detrimental than helpful and were wary of the promotion of this in a user guide: *“Teens liek to comapre, please don’t piush that agenda because it can harm mental health”* (W2).

The teens often disagreed on the role of competition in healthy self-tracking practices. Nonetheless, most participants felt the role of competition should be determined by one’s reasons for tracking. Some held a firm stance that everyday self-tracking should only be an individual activity without competition or comparison: *“That is why it’s called SELF tracking not compare yourself to others tracking”* (W4). On the other hand, participants acknowledged that, for teenagers tracking for clearly-defined reasons, such as athletes, competition can play an important role in their practices: *“i think that self tracking can be good for competition with peers, as in fitness goals and prepping for competitive sport”* (W5). The teens were keen for a user guide to offer guidance on both individual and healthy community self-tracking practices.

4.1.4 What Does it Mean? Importance of Context in Self-Track Data. Many teenagers expressed that numbers are not everything and they emphasized the importance of recognizing the context of self-tracked data. Participants believed quantitative data from self-tracking is not the only measure of wellbeing: *“Numbers don’t define you and don’t take the full picture into account”* (W4). This was one reason that some teens gave for moving away from competition and leaderboard-style data, as it often lacked this important context.

Participants felt the user guide should switch the framing from monitoring numbers alone to more dynamic goal setting which considers other measures of health and wellbeing. They wanted the guide to remind teens *“that it’s not all about the numbers humans aren’t numbers so don’t base you of that or define yourself of that”* (W4). Participants wanted others to be aware of how quantitative data is often limited and inaccurate. When describing self-tracked data, one participant said it *“Should always been taken with a pinch of salt, nothing is 100% accurate!”* (W4).

Participants focused on the holistic context of their data, rather than the numbers themselves: *“my weight and fitness level is the least interesting thing about me”* (W5). As such, they wanted the user guide to think beyond quantification and instead to offer guidance on how teens might explore more deeply the context of their self-tracked data. One participant described self-tracking goal setting by shifting their language use: *“if you don’t hit a goal, it does not mean that you’re a disappointment or you should give up, it means you have room to grow and become better”* (W5). Just as participants intentionally used cautionary language around self-tracking practices, they also wanted the user guide to purposefully use *“gentle”* (W5) language when considering wider contexts.

4.1.5 Who’s There? Balancing Privacy and the Need for Tailored Support. Participants wanted the user guide to address the role of third-party actors. The teens were critical of how their personal data was kept private, both by technology companies and within their social groups. There was widespread concern regarding the storage and security of their personal data. Participants wanted the user guide to answer questions such as: *“how is data stored”* (W4), *“is our data used for studies, do our devises save our data”* (W5), and *“how secure is the data”* (W4). Further, participants believed that their privacy should be respected and they sought guidance on how to keep their tracking private from peers and family if desired: *“i would love people to respect my privacy”* (W2).

Whilst the teens were wary of others accessing their private data, they wanted to be directed to information about third-party support. For example, many emphasized how the user guide must include resources for mental health support. Participants drew on their knowledge of other digital tools to make these suggestions: *“meetwo also has this section where you could reach out to third parties for support which is also nice”* (W2). These resources are not only for teenagers; participants emphasized the importance of improving parents’ understanding of the complex relationship between self-tracking and mental health, such as anxiety and body-image issues. As a participant in Workshop 1 warned, *“be careful who your buying them for because if it is a particularly anxious child then its best just to give them direct advice about excersise.”*

4.2 How Do Teenagers Think Information Should Be Presented in a Self-Tracking User Guide?

Participants wanted the user guide to be simple, with straightforward sentences or bullet points and *“gentle”* (W5), *“peaceful”* (W4) language. The teens wanted to leverage language to positively reframe the goals of self-tracking away from a focus on inadequacy: *“it should be presented as a way to improve how already amazing you are”* (W5). Further, they argued that it was important the user guide should employ cautionary language. As one participant suggested, *“it should viewed as a inovative product with a few cautions like anything from caffeine to perscription drugs”* (W1).

Participants disagreed on visual presentation of data and wanted the user guide to inform them about different ways of tracking their data. Often preference for the presentation of data and information depended on participants’ ideas around possible reasons for self-tracking. Some felt graphical representations of quantitative data were most useful, whereas others wanted an explanation of qualitative tracking options to be included in the user guide. As discussed in 3.1.4, the teens focused on the context of self-tracked data and some expressed that the guide should emphasize how quantitative data is insufficient in capturing this context: *“Numbers should be accompanied by a comment reflecting how you’re feeling in the moment eg 10km run but miserable or 5km run and happy”* (W4).

4.3 Where Would Teenagers Choose for a Self-Tracking User Guide to Be Distributed?

Participants were divided over the distribution methods of the user guide. Some participants had strong opinions on the best ways to share this information, while others believed the communication

strategy was less important than ensuring the information is available. As one participant said, *"there are different ways of passing the information it could be with a video or poster or whatever it doesn't mean it just about the information being passed"* (W1).

Some teens believed schools were the ideal places to present this information, though in small, interactive discussions, rather than large assemblies. Whether these discussions should be compulsory or not was controversial. One participant explained possible unintended effects of schools discussing this topic: *"I think school can be great for spreading awareness for these things BUT since we normally associate schools with our normal school subjects then learning about tracking may imply that it is also an obligatory subject and students may feel pressured to partake in it"* (W2). Discussing this topic in schools was seen as potentially encouraging self-tracking, which some teens were wary of: *"honestly i would say schools and parents shouldn't encourage tracking at all"* (W5). Conversely, some believed that whilst engaging in self-tracking practices should be optional, education around PI systems should be obligatory: *"i think it needs to be talking about full stop I know in my experience I have never had a conversation with someone about tracking"* (W4).

Most participants agreed that external experts should discuss self-tracking, not teachers: *"I think it should be an expert that comes into schools to talk about it. I wouldn't feel comfortable with a teacher commenting on how I ate/exercised"* (W4). Further, many felt teachers were not well-educated on the subject. As such, our teens proposed that having an informed expert would provide students with a safe environment to discuss the complex topics in the user guide.

Many felt social media was another helpful place for sharing the user guide, as it was accessible and comfortable for teenagers to use. Teens also appreciated how social media afforded anonymity in accessing this information. Social media was where many come across self-tracking for the first time, and as such, they felt it would likely have a large reach: *"I think also on social media, because that's probably where most people pick up ideas about self tracking"* (W5).

5 DISCUSSION

This study aimed to explore what teens would look for in the design of a user guide for those interested in self-tracking. Teenagers' use of self-tracking tools is widespread [23] and they are often skilled at using such tools [18]; however, our participants still felt ill-informed about many aspects of PI. Previous work demonstrates the great variability in teenagers' self-tracking practices [5] and our findings highlight the challenge of designing a teen-centered resource to capture the needs of such a diverse user group. We see contradictions in teenagers' design priorities and our participants identified these intricacies as well. To try to resolve these tensions would do a disservice to the complexity of the teen experience. Instead, we found that many of the participants' considerations in designing a user guide centered around two key questions, highlighting both the individual *and* social nature of self-tracking practices:

- Why am I self-tracking?
- What support systems and communities are in place around my self-tracking?

5.1 Core Reasons for Self-Tracking

Central to our teens' discussions was the recognition that different people might vary in their approaches to PI for health. Whilst

they see positive opportunities for self-tracking, teenagers also share concerns that self-tracking can be stressful or can promote unhealthy competition [5, 23]. The risk teenagers see is that beyond a certain threshold of frequency of use, or importance of data in one's life, one might become obsessed with self-tracking. Teenagers' concerns with crossing this threshold echo the potential risks of self-tracking previously studied, such as disordered eating behaviors [7, 12]. Although there seems to be little agreement on where this threshold falls—and indeed teenagers acknowledge that the threshold might differ according to specific contexts—there is agreement that such a threshold does exist and crossing it could be detrimental to teens' mental and physical health. In response, teenagers emphasize that a user guide should encourage individuals to be reflective about their reasons for tracking.

Teenagers express a desire to move to a new, more supportive language around self-tracking. This differs from the motivational or behavior change language often associated with self-tracking tools and associated resources designed for adults [13, 15]. Teens resist features such as leaderboards and negative feedback around failing to reach goals. Instead, they are eager to see a more holistic approach embedded in educational resources, as well as being commonplace in support systems and communities built around self-tracking. Future research should consider working with teens to explore how to effectively introduce this new language into new or existing PI systems and educational resources.

5.2 Opportunities for Social Support and Community-Building

The themes in our data emphasize that teenagers see the broader data ecosystem beyond the user and the PI tool. This includes intimate social relationships (such as friends and family), as well as third-party actors (such as corporate entities and mental health services). Teenagers highlight that *choice* is central both to their self-tracking practices and to the support and communities built around them. Privacy is paramount and teens seek guidance on how to maintain their autonomy in decisions around self-tracking and sharing data with others. This parallels previous HCI work with both teen [19] and novice users [21].

The social affordances of PI tools are important not only for competition and comparison, but also for connection with others [9, 24]. Teenagers are keen to highlight what they might expect from positive communities of care and are enthusiastic about resources around self-tracking tools that might enable healthy competition, and foster support and information-sharing between teens. Future work should examine opportunities to include mental health support in self-tracking educational resources, building out caring communities beyond teenagers' intimate social networks.

Teenagers stress the importance of balancing care and guidance with their freedom of choice in their self-tracking practices and education. We suggest that schools can offer important opportunities for building communities and support around self-tracking (e.g., by starting self-tracking after school clubs); however, thought should be given to separating this from curriculum-based teaching to emphasize teens' autonomy in choosing whether and how to self-track. Suggestions to achieve this include bringing in external experts to talk about health-related tracking, and moving

away from assembly-style lectures towards small group, interactive activities.

6 CONCLUSION AND FUTURE WORK

There remain important directions for future work in this area. Although our recruitment strategy aimed to include current users, previous users, and non-users of self-tracking tools, non-users were underrepresented in our sample. Future research should explore the needs of non-users and how educational resources might look different for this group.

Ultimately, this late-breaking work takes an important first step in understanding how we might design teen-centered resources to support PI practices. Drawing on teens' insights, we suggest an interactive user guide with multiple pathways might be best suited for teenagers to capture their diverse needs and uses. This guide might be available in accessible formats on social media, written in simple language, and contain video content. The user guide could be signposted through schools and scaffolded with interactive discussions with external self-tracking experts to help foster healthy communities. Our data demonstrates the challenge of embracing complexity in teens' self-tracking practices and suggests that keeping individual reasons for self-tracking and opportunities for social support at the center will enable researchers and designers to build resources appropriate for teens' positive use of PI tools for health and fitness.

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